Sarah J. Noe

South Carolina Institute for Archaeology and Anthropology
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Appointment

Assistant Research Professor
2026 South Carolina Institute for Archaeology and Anthology, University of South Carolina, Columbia SC

Education

Ph.D. in Anthropology, University of California, Santa Barbara, CA
2026 *Committee: Amber VanDerwarker, Sarah McClure, Doug Kennett, Greg Wilson and Randy Haas*

2017 M.A in Anthropology, University of Maryland, College Park, MD
Committee: George Hambrecht, Barnet Pavao–Zuckerman, and Michael Paolisso

2013 B.S. in Anthropology and B.A in English, Mercyhurst University, Erie, PA *summa cum laude*

Research Interests

- North American archaeology, with emphasis on Midwestern and Southeastern Mississippian communities
- Human–environment relationships, climate change, and resource management
- Zooarchaeology, vertebrate osteology, and human–animal relationships
- Foodways, subsistence, and Indigenous food systems
- Archaeological science: stable isotopes, radiocarbon and ancient DNA

Publications

*Journal cover art

[1] Noe, Sarah J., Haas R., and Aldenderfer M.,
(2025) **Hunting to herding on the Andean Altiplano: Zooarchaeological insights into Archaic Period subsistence in the Lake Titicaca Basin, Peru (9.0–3.5 ka)**. *Journal of Anthropological Archaeology*
https://doi.org/10.1016/j.jaa.2025.101658

[2] Noe, Sarah J., VanDerwarker A.M, Wilson G., Kuehn S., and Haas H..
(2024) **An Examination of Shifting Foodways during the Twelfth and Thirteenth Centuries in the Central Illinois River Valley**. *Southeastern Archaeology*
https://doi.org/10.1080/0734578X.2024.2420149

[3] Noe, Sarah J.
(2024) **A review of Cattle in the Postcolumbian Americas: A Zooarchaeological Historical Study** by Nicolas Delsol, Gainesville: University Press of Florida, 2024, xi + 266 pp. *Journal of Anthropological Research*
https://doi.org/10.1086/732444

[4] Noe, Sarah J., McCool W.C., Wilson, K.M.
(2024) **Cross–Regional Examination of Camelid Herding Practices in Peru from 600 BCE to 1450 CE: Insights from Stable Isotopes in Camelid Bone Collagen and Fiber.** *PloS one, 19(10), e0306205.*
https://doi.org/10.1371/journal.pone.0306205

[5] Byers, David A., Welker, M.H., Semanko, A., Itie, D., **Noe, S.J.,** Vasquez, J., Holland–Lulewicz, I. and McClure, S.B.
(2024) **Cranial variability in North American domestic and wild canids**. *Journal of Archaeological Science: Reports,* 57, 104590.
https://doi.org/10.1016/j.jasrep.2024.104590

[6] Noe, Sarah J., Wilson G., VanDerwarker A., George R.J., Kennett D.J.,
(2023) **Deer, Drought, and Warfare: Managing Risk in the Central Illinois River Valley (CIRV) from the Eleventh through Fourteenth century**. *Quaternary International, 689, 106–119.*
http://dx.doi.org/10.1016/j.quaint.2023.08.001

[7] *Noe, Sarah J.,
(2023) **Indigenous Foodways as Persistence in the Alta California Mission System**. *American Antiquity,* 88(4), 451–475.
http://dx.doi.org/10.1017/aaq.2023.53

[8] *Noe Sarah J.,
(2023) **Subsistence and Persistence: Indigenous Foodways within Mission Santa Clara de Asìs**. *California Archaeology, 15(1), 69–107.*
http://dx.doi.org/10.1080/1947461X.2023.2191239

[9] Hixon, S.W., Domic, A.I., Douglass, K.G., Roberts, P., Eccles, L., Buckley, M., Ivory, S., **Noe, S.J.** and Kennett, D.J
(2022). **Cutmarked bone of drought–tolerant extinct megafauna deposited with traces of fire, human foraging, and introduced animals in SW Madagascar**. *Scientific reports*, 12(1), 18504.
http://dx.doi.org/10.1038/s41598–022–22980–w

[10] Noe Sarah J.,
(2021) **Zooarchaeology of Mission Santa Clara de Asís: bone fragmentation, stew production, and commensality**. *International Journal of Historical Archaeology*, 26(4), 1–43.
http://dx.doi.org/10.1007/s10761–021–00632–5

[11] Haas R., Watson J., Buonasera T., Southon J., Chen J.C., **Noe S.J.**, Smith K., Llave C.V., Eerkens J. and Parker G.,
(2020) **Female hunters of the early Americas**. *Science Advances*, *6*(45), peabd0310.
http://dx.doi.org/10.1126/sciadv.abd0310

[12] Haas R., Buonasera T., Chen J., Eerkens J., **Noe S.J.,** Parker G., Smith, K., Viviano C. and Watson J.,
(2020) **Behavioral adaptations to life in the interior high Andes, 9–7ka**. In *American Journal of Physical Anthropology* 171:110–110

Manuscripts in Review

[13] Noe, Sarah J., Kracht E.C., Deagan B., and McCool W.M, (Accepted) **Assessing Starvation Signatures and Violence Through Nitrogen Isotopes in the LIP Nazca Highlands.** *American Journal of Biological Anthropology*

[14] Noe, Sarah J., VanDerwarker A.M, Wilson G., Bennaton S., and Vargas N. (Accepted) **Specialized white-tailed deer (Odocoileus virginianus) use at the early Mississippian Fandel site (11WD4) in the Central Illinois River Valley.** *American Antiquity*

[15] Haas, Randall, Allison Caine, Sean Field, Isabelle Guerrero, Sarah Noe, Jennifer Chen, Luis Flores-Blanco, Bryna Hull, Nathaniel R. Kitchel, Patricia J. McNeill, Katherine Morucci, Emmanue Oladeji, and Carlos Viviano Llave. (Under review). **Social complexity and the world's highest communal hunting traps in the Aymara Andes**. *Proceedings of the National Academy of Sciences*

[16] Noe, Sarah J., Haas R., Aldenderfer M., George R.J., and Kennett D.J., (Under Review) **Beyond Subsistence: The Role of Camelid Pastoralism in Andean Social and Political Complexity.** *Scientific Reports*

[17] Noe, Sarah J., Haas R., Aldenderfer M., George R.J., and Kennett D.J., (Submitted) **Human–Camelid Interactions during the Archaic–Formative Transition (ca. 5200–3000 cal BP): Stable Isotope Evidence from the Lake Titicaca Basin.** *Archaeological and Anthropological Sciences*

[18] Kracht, Emily, Noe, Sarah J., George R.J., and Kennett D.J., (Submitted) **Investigating Hutia Management in the Caribbean through Stable Isotope Analysis.** *Environmental Archaeology*

[19] Noe, Sarah J., Haas R., Aldenderfer M., George R.J., and Kennett D.J., (In prep) **From Deer to Fish: Resource Depression and Subsistence Change in Greater Cahokia.** *Journal of Anthropological Archaeology*

[20] Noe, Sarah J., Rademaker K., Moore K., Milton E., George R.J., and Kennett D.J., (In prep) **Ancient Camelid Mitogenomes from Cuncaicha and Panaulauca: Maternal Diversity from the Pleistocene to the Late Holocene.** *Nature Communications*

Funding and Awards

Research Funding

2026	UofSC Creative Research Grant	$27,000
2026	UofSC Equipment Grant	$33,000
2025	UCSB Academic Senate Committee Research Grant	$20,000
2025	Doctoral Dissertation Research Improvement Grant, NSF	$31,128
2024	Summer Research Block Grant	$3,500
2024	Rust Family Fellowship	$8,855
2023	Broom Center Research Block Grant	$2,500
2022	UCSB Social Science Research Accelerator Award	$5,500
2022	UCSB Summer Research Fellowship	$2,000
2022	UC ARP Summer Fellowship	$7,000

2021	Decolvenaere Research Accelerator Award	$500
2021	UCSB Anthropology Department Summer Research Grant	$2,000
2019	UC Summer Research Block Grant	$2,200
2019	Charles J. Erasmus Fund	$600
2018	UC Summer Research Block Grant	$2,000
2018	Lewis & Clark Fund for Exploration & Field Research	$5,000
2017	Phi Delta Gamma, UMD chapter Award for Excellence	$1,500
2016	University of Maryland Chair's Graduate Summer Research Fellowship	$2,500
2012	Mercyhurst Archaeological Institute Undergraduate Research Grant	$800

Fellowships and Funding

2019	Critical Mission Fellowship	$38,000
2017	National Science Foundation Graduate Research Fellowship	$138,000
2017	UCSB Anthropology Department Teaching Assistantship	$169.000
2016	University of Maryland Environmental Science Department Teaching Assistantship	$33,000
2015	University of Maryland Department of Anthropology Teaching Assistantship	$28,000

Awards

2024	Jamie Chad Brandon Student Paper Competition for the Annual Meeting of the Society for Historical Archaeology (SHA)	$500

Travel Funding

2023	Pacific Coast Archaeological Society Travel grant	$500
2022	GSA Conference Travel Grant	$800
2018	GSA Conference Travel Grant	$500
2016	University of Maryland Anthropology Department Travel Grant	$500
2012	Catherine McAuley Travel Scholarship, Mercyhurst University	$3,500

Teaching Experience

Teaching Interests

Introduction to Archaeology, Archaeological Science: Materials and Methods, Ancient Technologies: Production, Use, and Innovation, Zooarchaeology and Archaeological Faunal Analysis, Archaeological Chemistry: Isotopes, Genetics, and Materials, Latin American Archaeology, North American Archaeology, Environmental Archaeology and Human Adaptation, Sustainability and Heritage in Archaeology

Instructor of Record

University of South Carolina

ANTH 101 Introduction to Archaeology, Primates, People and Prehistory (Fall 2026)

ANTH 287 Ancient and Modern Food Production (Fall 2026)

University of California, Santa Barbara

ANTH 3 Introduction to Archaeology (Summer 2025)

ANTH 3 Introduction to Archaeology (Summer 2024)

ENV 149 Food, Agriculture, and the Environment (Summer 2024)

ANTH 3 Introduction to Archaeology (Summer 2023)

ENV 149 Food, Agriculture, and the Environment (Summer 2023)

Teaching Assistant

University of California, Santa Barbara

ANTH 3 Introduction to Archaeology (Spring 2025)

ANTH 5 Introductory Biological Anthropology (Winter 2025)

ANTH 3 Introduction to Archaeology (Fall 2024)

ANTH 3 Introduction to Archaeology (Spring 2024)

ANTH 133 Cultural Development of Mesoamerica (Winter 2024)

ANTH 3 Introduction to Archaeology (Fall 2023)

ANTH 3 Introduction to Archaeology (Spring 2023)

ANTH 133 Cultural Development of Mesoamerica (Winter 2023)

ANTH 3 Introduction to Archaeology (Fall 2022)

ANTH 3 Introduction to Archaeology (Summer Session A 2021)

ANTH 131 California Indians (Winter 2022)

ANTH 3 Introduction to Archaeology (Fall 2021)

ANTH 109 Human Universals (Summer Session B 2020)

ANTH 3 Introduction to Archaeology (Summer Session A 2020)

ANTH 5 Introductory Biological Anthropology (Spring 2020)

ANTH 131 California Indians (Winter 2020)

ANTH 3 Introduction to Archaeology (Fall 2019)

University of Maryland, College Park

ENSP101 Introduction to Environmental Science

ENSP102 Introduction to Environmental Policy

ANTH240 Introduction to Archaeology

ANTH260 Introduction to Socio–cultural Anthropology and Linguistics

Workshop Instructor

UCSB 4th Annual Workshop Graduate Students

2026 R Programming for Social Scientists

UCSB 5th Annual Workshop Graduate Students

2024 Intro to Data Analysis and Visualization in Python for Social Scientists

Mentorship

University of South Carolina

- Advise four student researchers: three M.A. students and one undergraduate.
- Mentor one undergraduate researcher in developing a Magellan Scholar proposal.
- Mentor one M.A. student and one undergraduate researcher on a public-facing historical archaeology project.

University of California, Santa Barbara

- Advised eight undergraduate researchers on independent research projects.
- Mentored seven undergraduate and two M.A. researchers in research methods and analytical skills through participation in my research.
- Provided guidance on components of four M.A. theses; all four students successfully defended.

Invited Lectures

Public Talks

2026 *Food Insecurity, What We Can Learn From the Past*
Native American Studies Center, Lancaster, SC,

2025 *Ancient Foodways, Modern Problems*
Dixon Mounds Museum, Illinois
Santa Barabra Museum of Natural History

2023 *Foodways and Persistence: Native American Diet at Mission Santa Clara(1777–1836)*
Santa Barabra Museum of Natural History

2021 *Zooarchaeological Insights into the Persistence of Native American Diet at Mission Santa Clara de Asís (1777–1836)*
Santa Barabara Museum of Natural History

2019 *Risk, Agriculture, and Climate Change: Sustainable Viking Foodways System*
UC Santa Barbara Colloquium

2017 *Archaeology and Climate Change: Interdisciplinary Studies and Talking Across the Aisle*
University of Maryland, College Park Library Series: Interdisciplinary Dialogues

2014 *You Are What You Eat: What Archaeology Adds to Understanding Foodways*
expERIEnce Children's Museum, Erie PA

Invited Academic Lectures

Dickinson College

2022 *Foodways and Climate Change: Examination of Native American Diet at Mission Santa Clara (1777–1836)* Environmental Archaeology (undergraduate course)

2021 *Sustainable Foodways in the Face of Climate Change in the Early Modern Period* Environmental Archaeology (undergraduate course)

Northern Illinois University

2021 *Introduction to Zooarchaeology* Exploring Archaeology (undergraduate course)

2022 *Introduction to Zooarchaeology* Exploring Archaeology (undergraduate course)

University of California, Santa Barbara

2025 *Understanding Paleoecology* Introduction to Archaeology (undergraduate course)

2020 *Sustainable and Ancient Food Production in the Andes* Introduction to Archaeology (undergraduate course)

2018 *Food Security and Climate Change* Introduction to Archaeology (undergraduate course)

University of Maryland, College Park

2017 *Climate Change and the Longue Durée* Environmental Science (undergraduate course)

2016 *Food, Climate Change, and the Legacy of Colonialism* Introduction to Environmental Policy (undergraduate course)

Field & Research Experience

2026	**Historic Horseshoe Archaeological Field School — University of South Carolina, Columbia, South Carolina** Helped lead an on-campus field school investigating the Historic Horseshoe in collaboration with local descendant communities. Trained students in archaeological excavation and documentation methods.
Summer 2026	**Carabamba Archaeological Excavations and Field School — Northern Highlands, Peru** Led excavations and trained field school students in excavation, documentation, and archaeological recovery methods. Conducted zooarchaeological analysis of camelid remains to investigate herd management and food systems.
August 2026–Present	**Mississippian Faunal Analysis — Tombigbee River Valley, Mississippi** Conduct zooarchaeological analysis of faunal remains to investigate hunting, fishing, and animal resource use in Mississippian food systems.

September 2025–Present	**Camelid Population History in the Central Andes — Junín Valley, Peru** Collaborate with Kurt Rademaker on archaeological camelid remains from the central Andes. Analyze ancient mitochondrial genomes to reconstruct maternal diversity from the Pleistocene through the Late Holocene and investigate lineage persistence across the transition from hunting to herding.
September 2023–Present	**Camelid Domestication in the Lake Titicaca Basin — Peru** Integrate ancient mitochondrial genome sequencing with AMS radiocarbon dating and stable isotope sample preparation and analysis to investigate camelid domestication and population history.
September 2023–Present	**Mississippian Beginnings: Foodways and Climate Change — Greater Cahokia and Central Illinois River Valley, Illinois** Conduct zooarchaeological analysis of faunal assemblages to investigate changes in hunting, fishing, and resource management during the development of Mississippian communities and the Medieval Climatic Anomaly.
Summer Field Seasons, 2019–2025	**Mississippian Beginnings: Excavations at Fandel — Central Illinois River Valley, Illinois** Led excavations of residential structures adjacent to platform mounds at Fandel. Investigated Mississippian settlement organization and recovered materials for research on foodways and human–environment relationships.
2021–2024	**Living with Warfare — Illinois River Valley, Illinois** Prepared deer bone samples and conducted stable carbon and nitrogen isotope analysis for six archaeological sites to assess hunting practices during periods of climate change and warfare.
June–August 2019	**Camelid Herding in the Lake Titicaca Basin — Puno, Peru** Conducted zooarchaeological analysis of faunal remains from three Archaic Period sites at the Collasuyo Archaeological Research Institute (CARI).
June–July 2018	**Religion and Power from a Household Perspective at Cerro Tortolita — Peru** Served as crew chief for excavations of residential structures adjacent to the ceremonial center at Cerro Tortolita.
January 2016–May 2017	**Comparative Island Ecodynamics in the North Atlantic Project — Iceland** Conducted zooarchaeological analysis of faunal remains from Skálholt, the residence of the Bishop of Iceland.
2015	**Hardy Park Site Archaeological Investigations — Moorefield, West Virginia** Served as crew chief and team member for Phase III excavations of a Woodland-period Native American site with Stell Environmental, Virginia Beach, Virginia.
June–July 2014	**The First Americans: Excavations at Meadowcroft Rockshelter — Pennsylvania** Participated in Phase III excavations of a new quadrant at Meadowcroft Rockshelter under Principal Investigator James Adovasio.

June–August 201	**Taos Collaborative Archaeological Project (TCAP) — Taos, New Mexico** Served as crew chief and team member for Phase III excavations at Fort Burgwin under Principal Investigators Judith Thomas and James Adovasio.

Work Experience

Sept 2015–Dec 2017	*Center for Global Sustainability/Integration and Application Network, University of Maryland, College Park, MA* Collaborated with federal agencies and academic institutions to advance climate change mitigation and sustainability initiatives. Performed ecological assessments and data analysis, assisted in using visually engaging icons to distill research findings and ensure data accessibility across cultural and language barriers.
Sept 2014–Dec 2015	*Environmental Defense Fund, Boston MA* Collaborated with a dedicated team to implement a cost–effective electronic monitoring program that accurately tracks commercial groundfish activities in the Gulf of Maine. Analyzed historical and current cod stock data to identify critical factors such as overfishing, poor management practices, and climate change impacts that led to the decline of cod populations to 5–8% of sustainable levels. Worked closely with government agencies, fishermen, and stakeholders.
Jun 2013–May 2014	*Earth Day Network, Washington D.C* Contributed to an initiative aimed at reducing methane emissions from the agricultural sector. Working collaboratively with top food companies, farmer organizations, and policymakers, our team implemented sustainable methane reduction strategies that supported both environmental goals and economic viability. I was actively involved in developing and promoting regional solutions tailored to specific agricultural practices.

Professional Meetings

Organized Symposia

[1] *Interdisciplinary Approaches In Zooarchaeology: Addressing Big Questions with Ancient Animals* (2024) Sponsored by Zooarchaeological Interest Group), Society for American Archaeology, New Orleans. Co–Organized with Hugh Radde

[2] *Symposium: From Interactions to Management: Mutualistic Human–Animal Relationships in the Archaeological Record of the Americas* (2026) Sponsored by Zooarchaeological Interest Group), Society for American Archaeology, New Orleans. Co–Organized with Daniel Dalmas

Conference Presentations

[1] Noe Sarah J, Randy Hass, Mark Aldenderfer
(2024) From Hunting to Herding in the Lake Titicaca Basin: An Investigation of Faunal Assemblages, 9.0—3.5 ka. Paper presented at the Society for American Archaeology, Denver, CO

[2] Noe Sarah J.
(2024) Foodways within the Alta California Mission System: Assessing Colonial and Indigenous Diet within Mission Santa Clara de Asís. Paper presented Historical and Underwater Archaeology, Boston, MA.

[3] Noe Sarah J., Greg Wilson, Steve Kuehn
(2023) Faunal Patterns at Cahokia and the CIRV: A Comprehensive Review. Paper presented at the Southeastern Archaeological Conference, Chattanooga TN

[4] Noe Sarah J., Greg Wilson, Amber VanDerwarker, Richard George, Doug Kennett
(2023) Deer, Drought, and Warfare: An isotopic investigation of shifts in deer hunting from the eleventh through the fourteenth centuries in the Central Illinois River Valley (CIRV). Paper presented at the Society for American Archaeology, Portland, OR

[5] Noe Sarah
(2023) Situating Spanish and Indigenous Diet and Foodways within Mission Santa Clara. Paper presented at the Society for California Archaeology, Oakland CA

[6] Noe Sarah, Amber VanDerwarker and Greg Wilson
(2022) An Examination of Shifting Foodways during the Twelfth and Thirteenth Centuries in the Central Illinois River Valley Paper presented at the Society for American Archaeology, Chicago, Il.

[7] Noe, Sarah
(2021) Change, Continuity and Foodways: Indigenous Diet at Mission Santa Clara (1777–1836). Paper presented at the Society for American Archaeology, online

[8] Noe, Sarah
(2021) Zooarchaeological Insights into Native American Diet at Mission Santa Clara (1777–1836). Paper presented at Society for California Archaeology, online

[9] Noe, Sarah, Randy Haas, Mark Aldenderfer
(2020) *From Hunting to Herding: A Preliminary Investigation of Camelid Domestication in the Lake Titicaca Basin.* Paper was to be presented at the 85th annual meeting of the Society for American Archaeology, Austin, TX.

[10] Haas, Randall, Tammy Buonasera, Jennifer Chen, Jelmer Eerkens, Sarah Noe, Glendon Parker, John Southon, Carlos Viviano Llave, and James Watson
(2020) Behavioral Adaptations to Life in the Interior High Andes, 9–7ka. Paper was to be presented at the American Association of Physical Anthropologists, Los Angeles, CA.

[11] Noe, Sarah
(2020) Zooarchaeological Insights into Native American Diet at Mission Santa Clara (1777–1836). Paper was to be presented at the 54th annual meeting of the Society for California Archaeology, Riverside, CA.

[12] Haas, Randall, Tammy Buonasera, Jennifer Chen, Jelmer Eerkens Sarah Noe, Glendon Parker, John Southon, Carlos Viviano Llave, and James Watson.
(2020) *Excavations at Wilamaya Patjxa, Lake Titicaca Basin, 9–7ka*. Paper presented at the Annual Meeting of the Institute of Andean Studies, University of California, Berkeley

[14] Noe, Sarah
(2020) *Icelandic Resilience: Food and Agriculture in the face of Climate Change*. Paper presented Historical Archaeology Conference, Washington DC.

Mentorship and Synergistic Activities

2019–2026	*Science on Tap — Draughtsmen Aleworks, UC Santa Barbara* Organized a public outreach series featuring researchers from across disciplines presenting their work through accessible talks and audience discussions in an informal brewery setting.
2024	*Alpha Sigma Kappa – Women in STEM Research Night*
2017 –2024	*Anthropology Graduate Student Association (AGSA), UCSB*
2019 –2024	*Board Member of WiSE (Womxn in Stem), UCSB* Assisted in planning monthly workshops aimed at connecting women in graduate school across departments for collaboration and support
2023 –2024	*Lead Organizer for HER Science outreach program* Assisted in organizing a summer program for high school students at the University of California, Santa Barbara, where they participated in lab activities on the weekend. The program also paired mentors with high school students to guide them through the college application process.
2020 –2024	*Assistant Organizer "STEM Forward" first–generation summer workshop* Created a mentorship network for first generation student at the University of California, Santa Barbara, pairing them with a graduate mentor to assist in their academic goals
2019–2021	*Mentor – Research Mentorship Program* Summer program for high school students to gain interdisciplinary, hands–on, university–level research.
2019	*FieldWORKshop: Navigating Challenges in Fieldwork, UCSB*
2017–2020	*STEM Squad: Women in STEM science outreach platform* Member of an outreach platform, worked with first generation students at UCSB.
2017–2018	*Representative – Graduate Student Association (GSA)*
2018 –2023	*AGSA Undergraduate and Graduate Student Mentorship Program, UCSB* Provides academic and professional support for undergraduate and graduate students in Anthropology and other departments.
2015 –2017	*Terp Young Scholars* Three weeks program during the summer at the University of Maryland for first generation students to get acclimated with college and familiar with an academic interest (in–person and online)